

# Agenda

## 1. Announcements

- I will have office hours today: 12:00-1:30 in VV B235
- HW 11.8 and 11.9 are due next week
- no discussion quiz this week

## 2.) Section 11.10

## Maclaurin Series

How to turn a polynomial into a power series?

Example:  $f(x) = \underbrace{-1}_{a_0} + \underbrace{2x}_{a_1} - \underbrace{5x^2}_{a_2} + \underbrace{6x^3}_{a_3} + 0x^4 + 0x^5 + \dots = \sum_{n=0}^{\infty} a_n x^n$

Can we come up with a formula that will always work? yes.

notice:

|                                |                            |                                                 |
|--------------------------------|----------------------------|-------------------------------------------------|
| $f(x) = -1 + 2x - 5x^2 + 6x^3$ | $f(0) = -1$                | $a_0 = f(0) = \frac{f(0)}{0!}$                  |
| $f'(x) = 2 - 10x + 18x^2$      | $f'(0) = 2$                | $a_1 = f'(0) = \frac{f'(0)}{1!}$                |
| $f''(x) = -10 + 36x$           | $f''(0) = -10$             | $a_2 = \frac{f''(0)}{2!} = \frac{f(0)}{2!}$     |
| $f^{(3)}(x) = 36$              | $f^{(3)}(0) = 36$          | $a_3 = \frac{f^{(3)}(0)}{3!} = \frac{f(0)}{3!}$ |
| $\vdots$                       | $\vdots$                   | $\vdots$                                        |
| $f^{(n)}(x) = 0, n \geq 4$     | $f^{(n)}(0) = 0, n \geq 4$ | $a_n = 0, n \geq 4$                             |
|                                |                            | $a_n = \frac{f^{(n)}(0)}{n!}$                   |

Theorem:

Telling us we can think of

$$f(x) = -1 + 2x - 5x^2 + 6x^3 = \frac{-1}{0!} + \frac{2}{1!}x - \frac{10}{2!}x^2 + \frac{36}{3!}x^3$$

Can generalize this, and extend it to a Maclaurin series

→ If a function  $f(x)$  has a power series representation centered at  $x=0$ , then  $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$ .

This power series is called the Maclaurin series  
(or the Taylor series centered at zero)

Example: Find the Maclaurin Series for  $f(x) = e^x$ . Then find its radius of convergence and interval of convergence.

$$f(x) = e^x = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

Take enough derivatives to recognize a pattern

①

$$\begin{array}{l|l} \begin{array}{l} f(x) = e^x \\ f'(x) = e^x \\ \vdots \\ f^{(n)}(x) = e^x \end{array} & \begin{array}{l} \text{centered at } 0, \text{ plugging in } 0 \\ f(0) = 1 \\ f'(0) = 1 \\ \vdots \\ f^{(n)}(0) = 1 \end{array} \end{array} \quad e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$$

② Find radius & interval of convergence

Ratio test

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} \right| = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0 < 1 \quad \text{always converges}$$

|                               |                                      |
|-------------------------------|--------------------------------------|
| $\underline{ROC}: R = \infty$ | $\underline{IOC}: (-\infty, \infty)$ |
|-------------------------------|--------------------------------------|

Activity: Find the Maclaurin series for  $f(x) = \sin x$ .

Then find its radius of convergence and interval of convergence.

$$f(x) = \sin x \quad f(0) = 0$$

$$f'(x) = \cos x \quad f'(0) = 1$$

$$f''(x) = -\sin x \quad f''(0) = 0$$

$$f'''(x) = -\cos x \quad f'''(0) = -1$$

$$f^{(4)}(x) = \sin x \quad f^{(4)}(0) = 1$$

$$\sin x = 0 + \frac{1}{1!}x + 0 + \frac{-1}{3!}x^3 + 0 + \frac{1}{5!}x^5 + \dots$$

$$\sin x = \frac{1}{1!} + \frac{-1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} x^{2n+3}}{(2(n+1)+1)!} \cdot \frac{(2n+1)!}{(-1)^n x^{2n+1}} \right|$$

$$= \lim_{n \rightarrow \infty} \frac{x^2}{(2n+3)(2n+2)} = 0 < 1 \quad \text{always converges}$$

ROC:  $R = \infty$

IOL:  $(-\infty, \infty)$

## Taylor Series

Theorem If  $f(x)$  has a power series representation centered at  $x=c$ , then

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n$$

This power series is called the Taylor series centered at  $x=c$ !

Example: Find the Taylor Series for  $f(x) = \frac{1}{x}$  centered at  $x=1$

$$f(x) = \frac{1}{x} \quad f(1) = 1 = 0!$$

$$f'(x) = -\frac{1}{x^2} \quad f'(1) = -1 = -1!$$

$$f''(x) = \frac{2}{x^3} \quad f''(1) = 2 = 2!$$

$$f'''(x) = -\frac{6}{x^4} \quad f'''(1) = -6 = -3!$$

$$f^{(n)}(x) = \frac{(-1)^n n!}{x^{n+1}}$$

$$f^{(n)}(1) = (-1)^n n! \quad \frac{f^{(n)}(1)}{n!} = (-1)^n$$

$$\frac{1}{x} = \sum_{n=0}^{\infty} (-1)^n (x-1)^n$$

Activity: Find the Taylor Series for  $h(x) = e^x$  centered at  $x = \ln 2$

$$h^{(n)}(x) = e^x \quad h^{(n)}(\ln 2) = e^{\ln 2} = 2 \quad \frac{h^{(n)}(\ln 2)}{n!} = \frac{2}{n!}$$

$$e^x = \sum_{n=0}^{\infty} \frac{2}{n!} (x - \ln 2)^n$$